Efficient Inference: A Deeper Look into Quantization

May 4, 2026

Recap: Representing numbers with float32

Key facts

8 bit biased exponent $\rightarrow \approx [10^{-38}, 10^{38}]$

24 bit fraction $\rightarrow$ **7 digit precision**

Recap: bfloat16

Sign (1 bit) **Exponent** (8 bits) **Fraction** (7 bits)

Key facts

8 bit biased exponent $\rightarrow \approx [\mathbf{10^{-38}, 10^{38}}]$ **Same as float32**

7 bit fraction $\rightarrow$ **3 digit precision**

Recap: int8

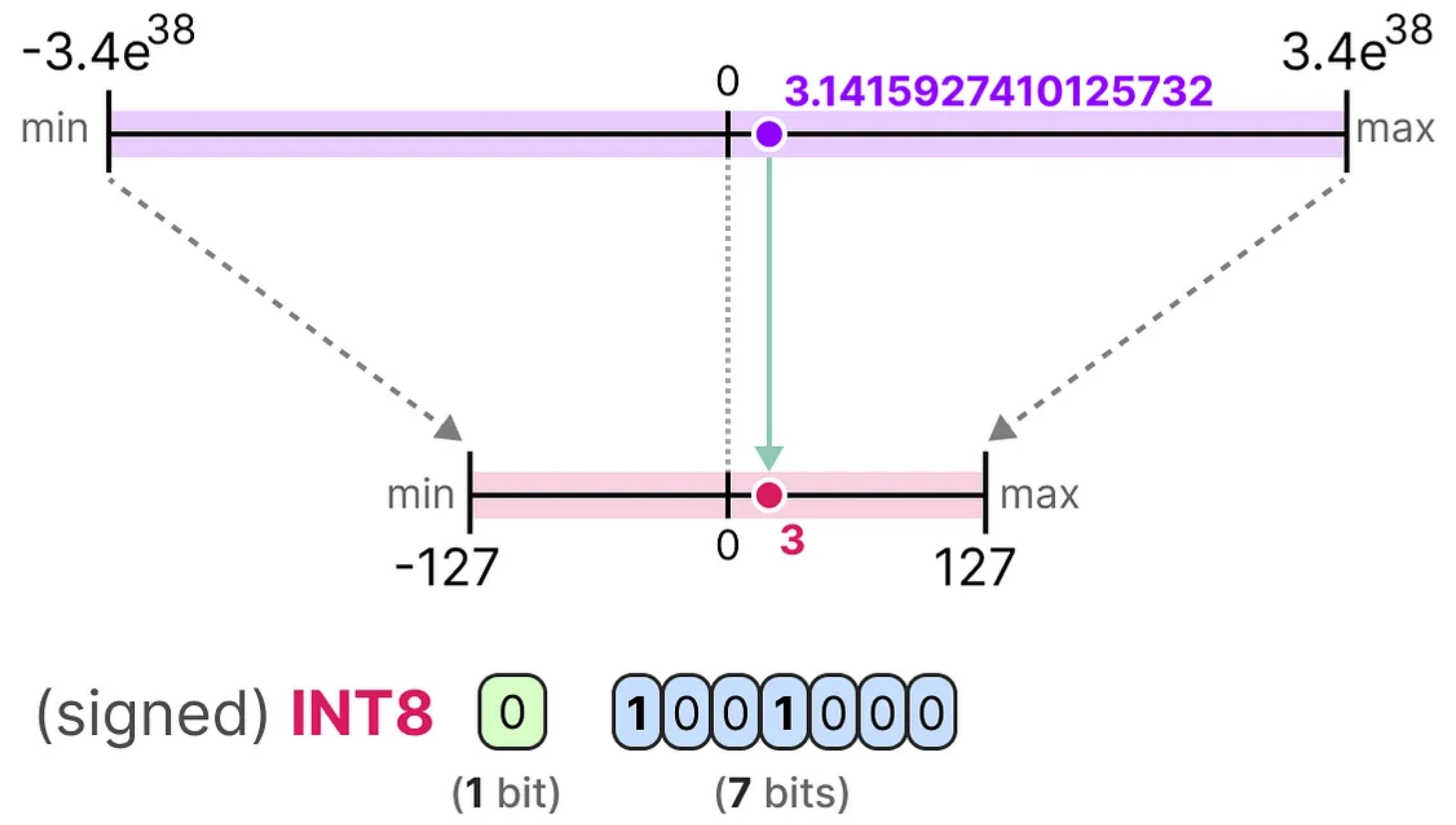
- Can represent 2^8 -1 = 255 values
- Range from [-128, 127]

Key facts

7 bits $\rightarrow [-128, 127]$

255 values only

Recap: Quantizing weights to int8



- **Quantization:** $\mathbf{W}_q = \text{round}\left(\gamma \cdot \mathbf{W}\right)$ with $\gamma = \dfrac{2^{N-1}}{\max(|\mathbf{W}|)}$ and N=8
- **Dequantization:** $\hat{\mathbf{W}} = \dfrac{\mathbf{W}_q}{\gamma}$

[Grootendorst]

This lecture

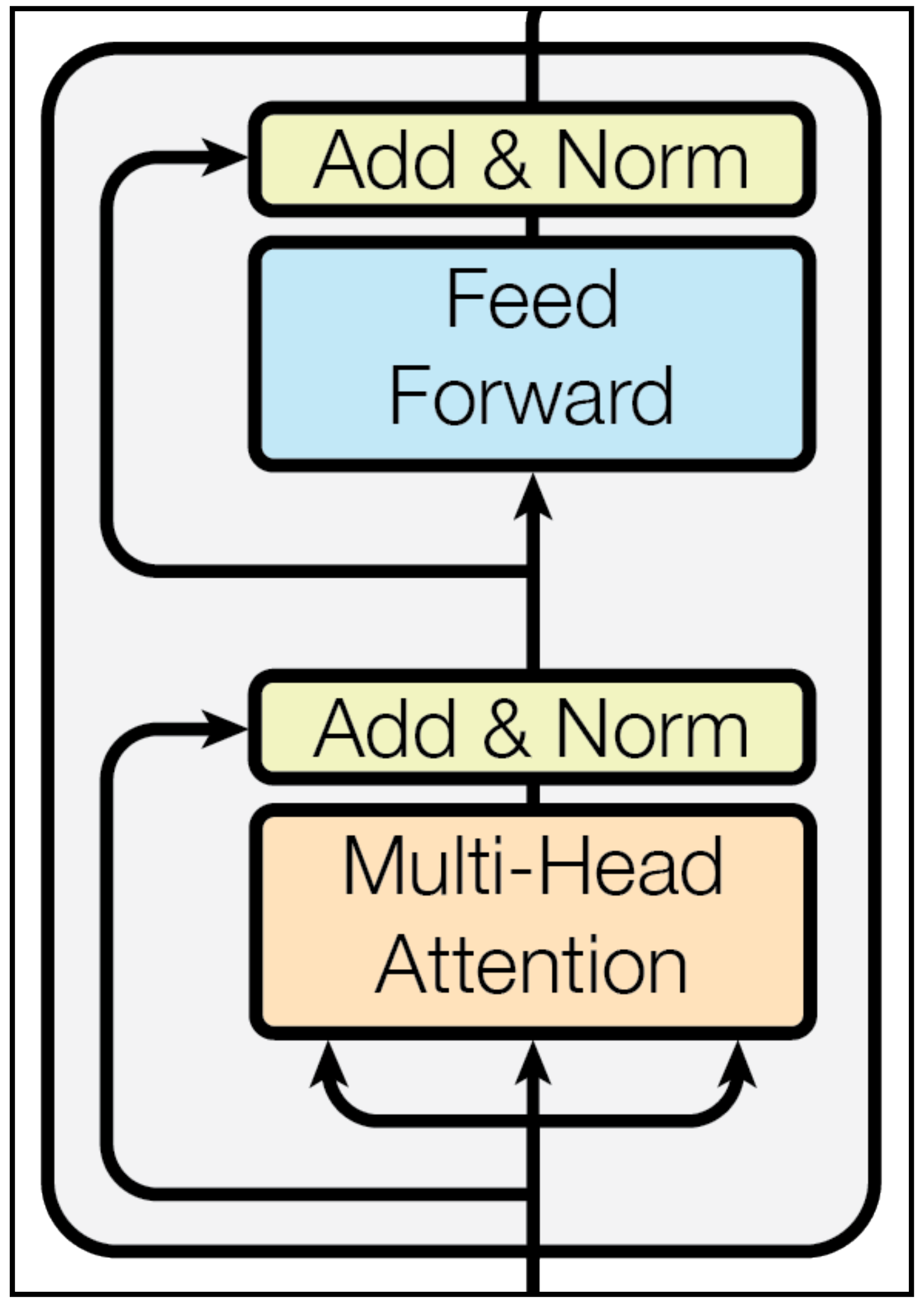
Explore various quantization techniques and learn about their pros and cons

A closer look at model parameters

- Most computations within the model consist of

$$\boxed{\mathbf{Y = X\,W + b}}$$

- Let us ignore the bias (B) computation
- Consist of much smaller portion of the computation
  - $O(D^2)$ operations in **XW**
  - $O(D)$ operations in **b**
- Biases often not quantized



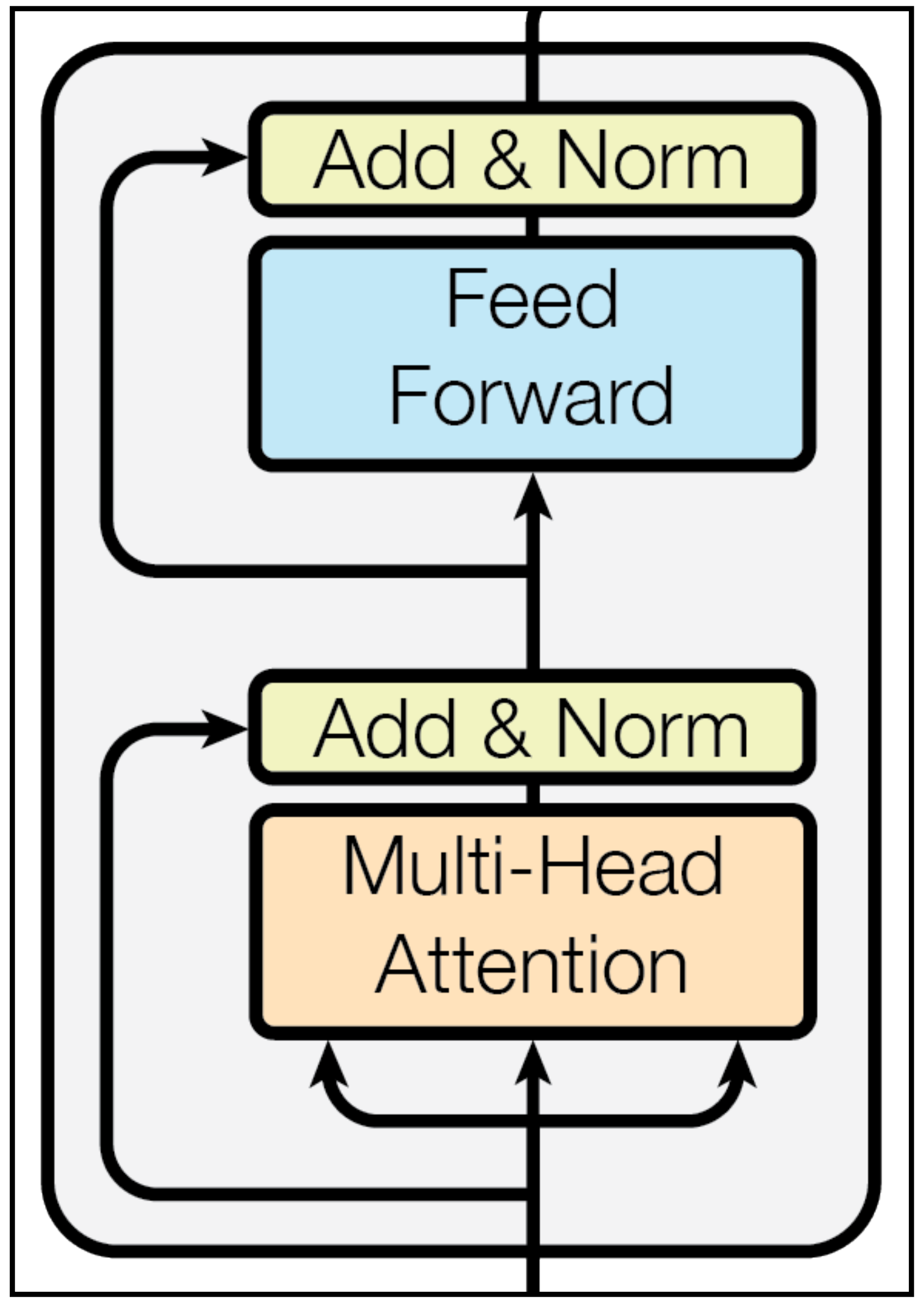
Vaswani et al.

A closer look at model parameters

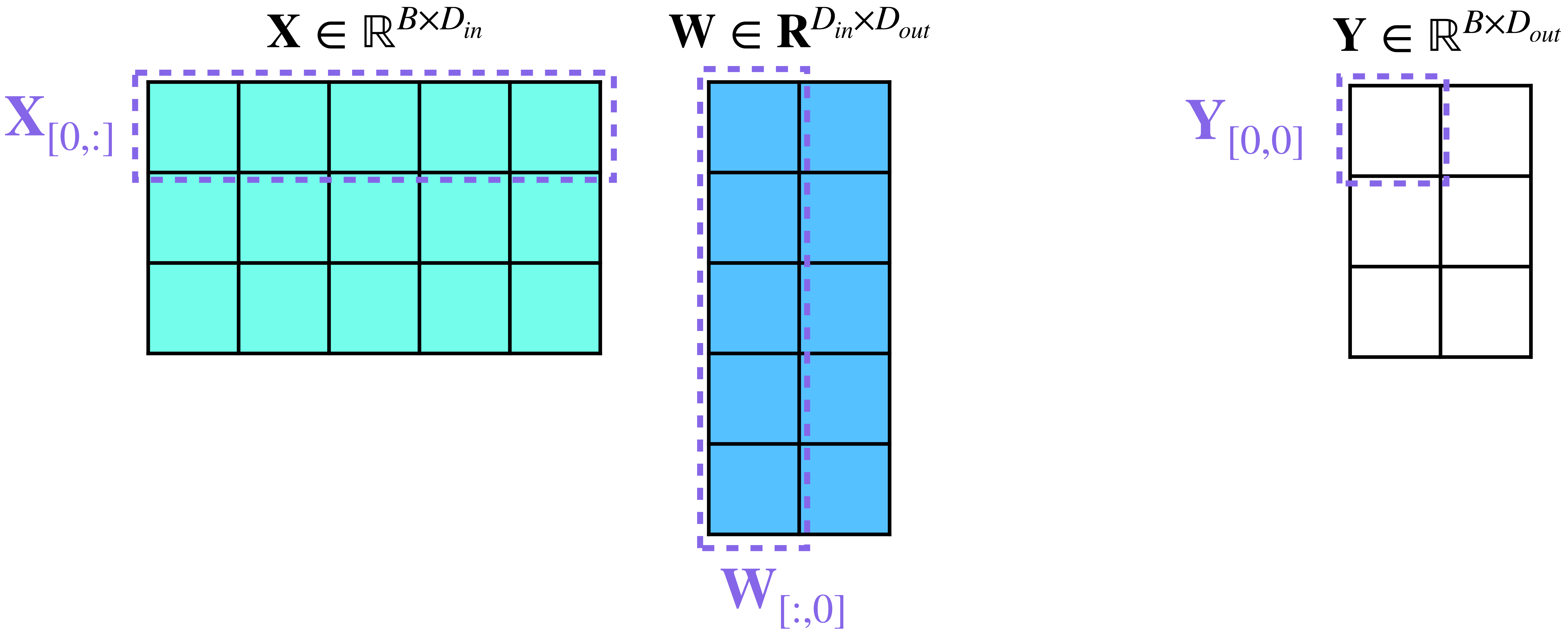
- Most computations within the model consist of

$$\boxed{\mathbf{Y} = \mathbf{X}\,\mathbf{W}}$$

- $\mathbf{X} \in \mathbb{R}^{B \times D_{in}}$ where B is the batch size
- Number of tokens could be another tensor dimensions
  - Will assume it to be 1 unless specified otherwise
- $\mathbf{W} \in \mathbf{R}^{D_{in} \times D_{out}}$
- The computation **XW** projects an input from D_{in} to D_{out} dimensions



A recap of matrix multiplication



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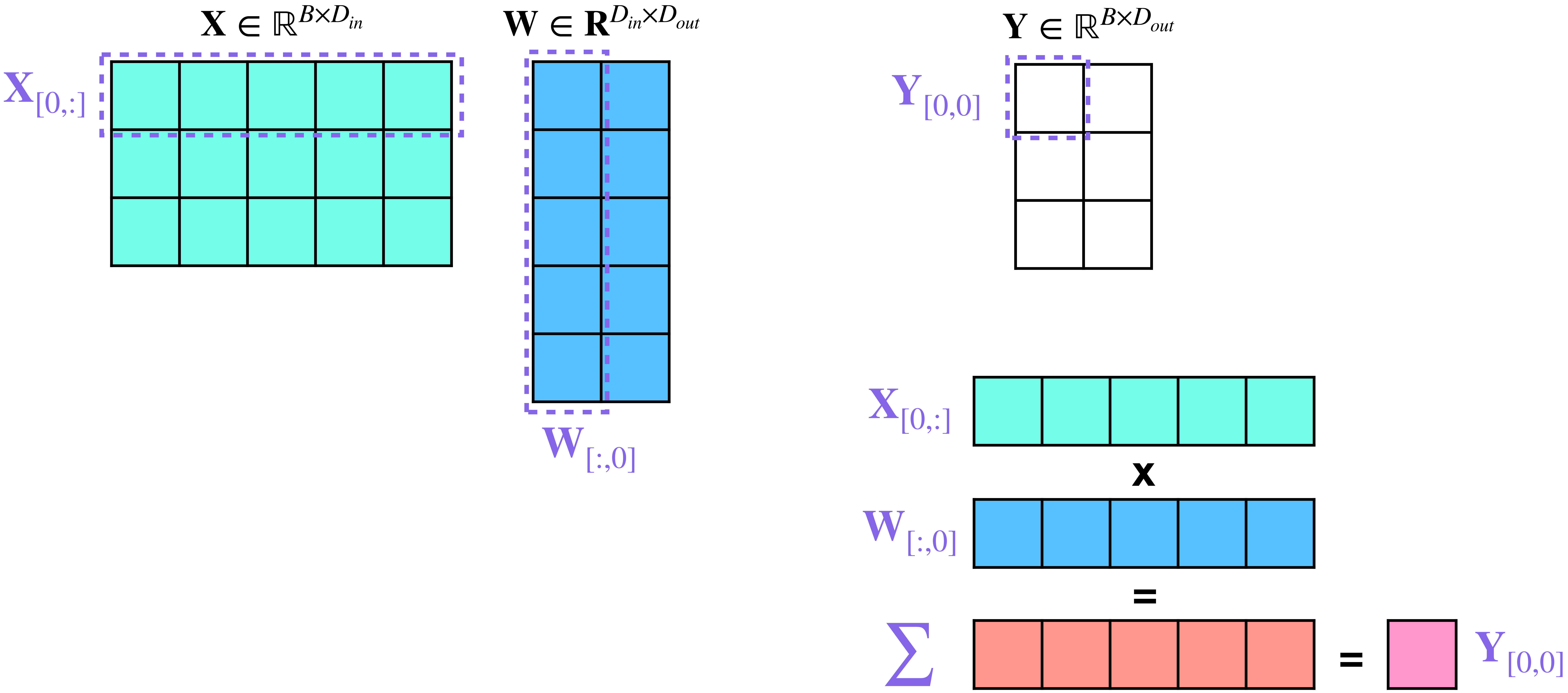

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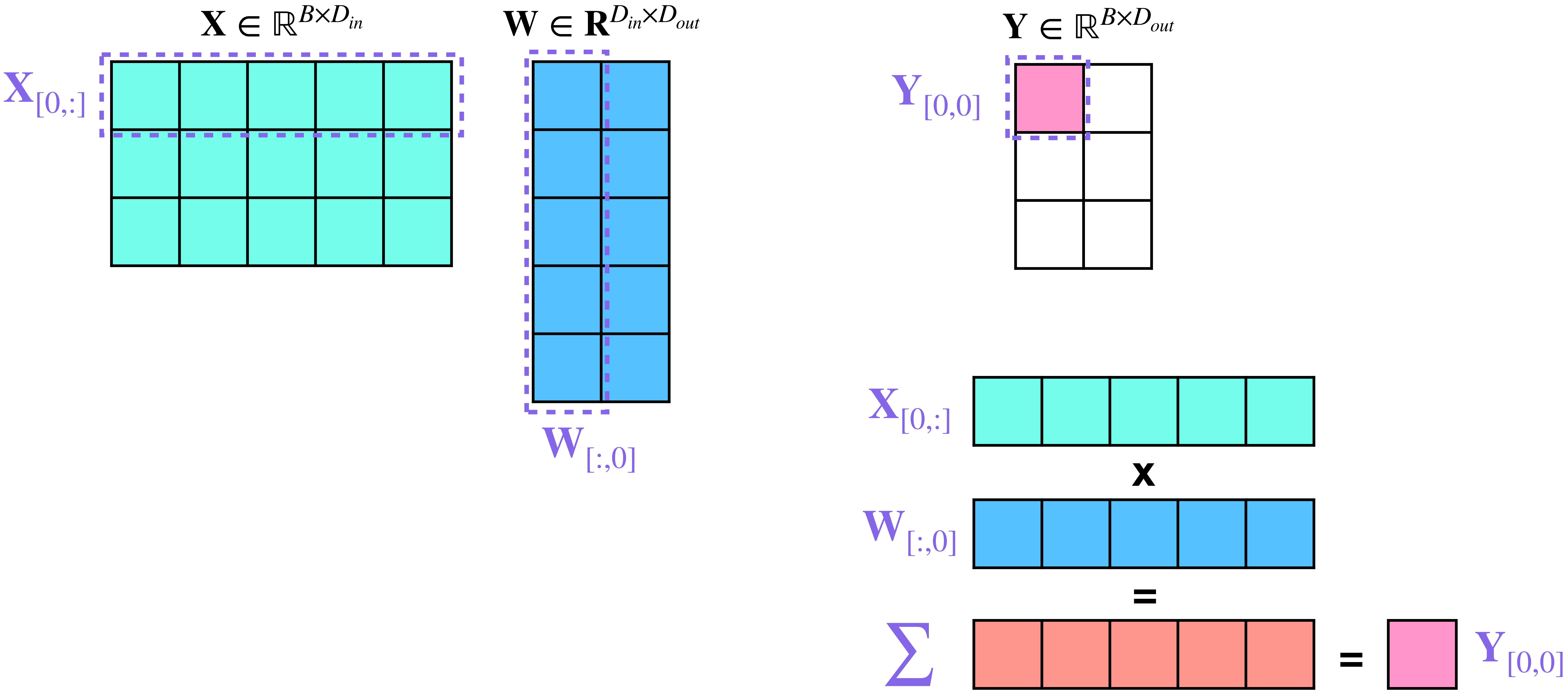

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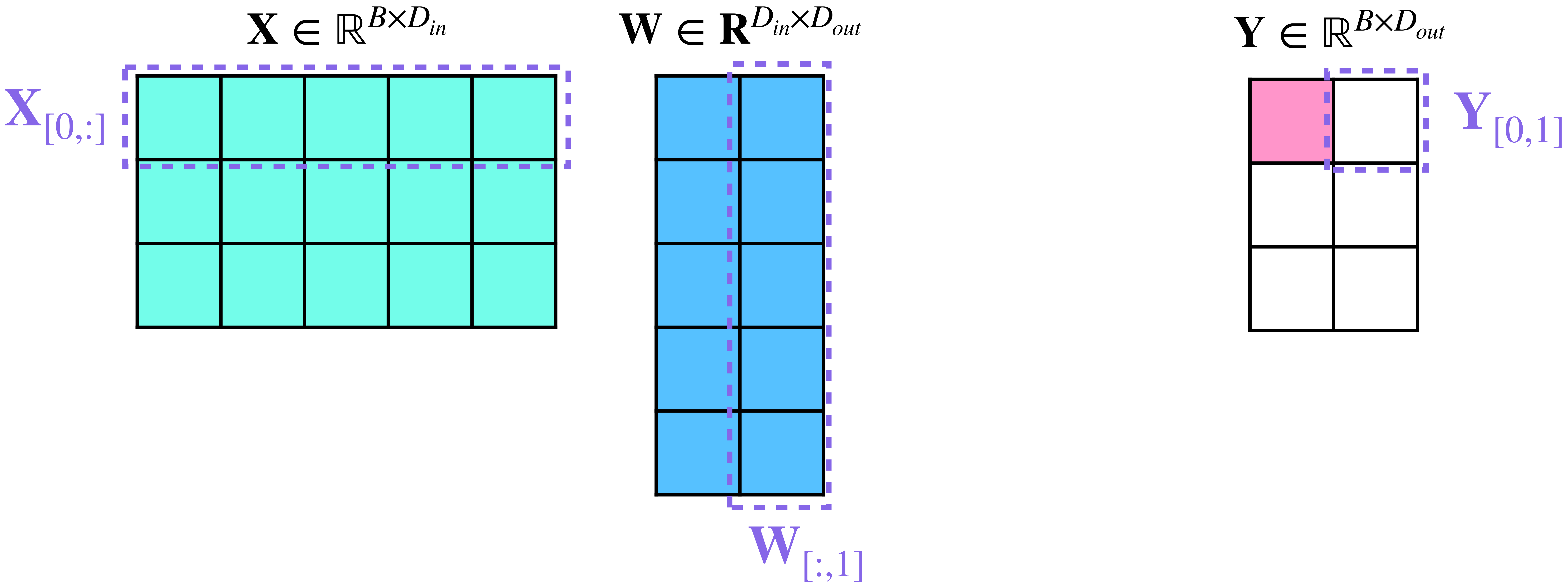

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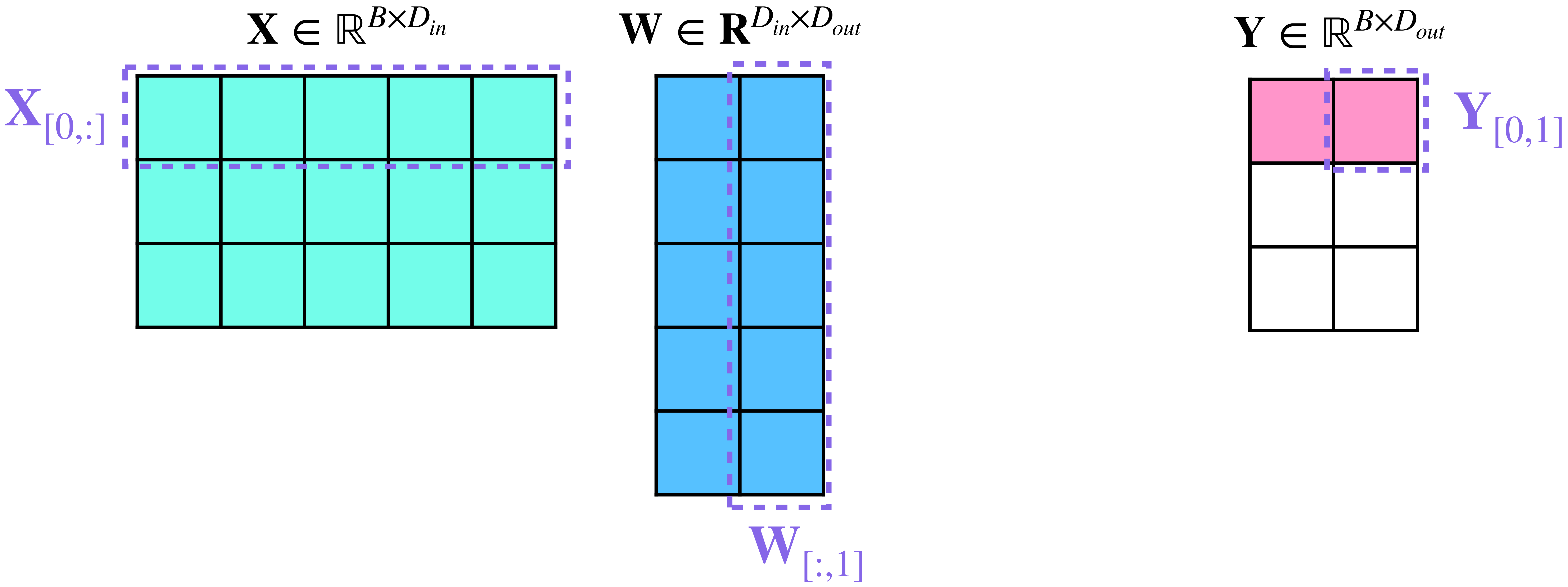

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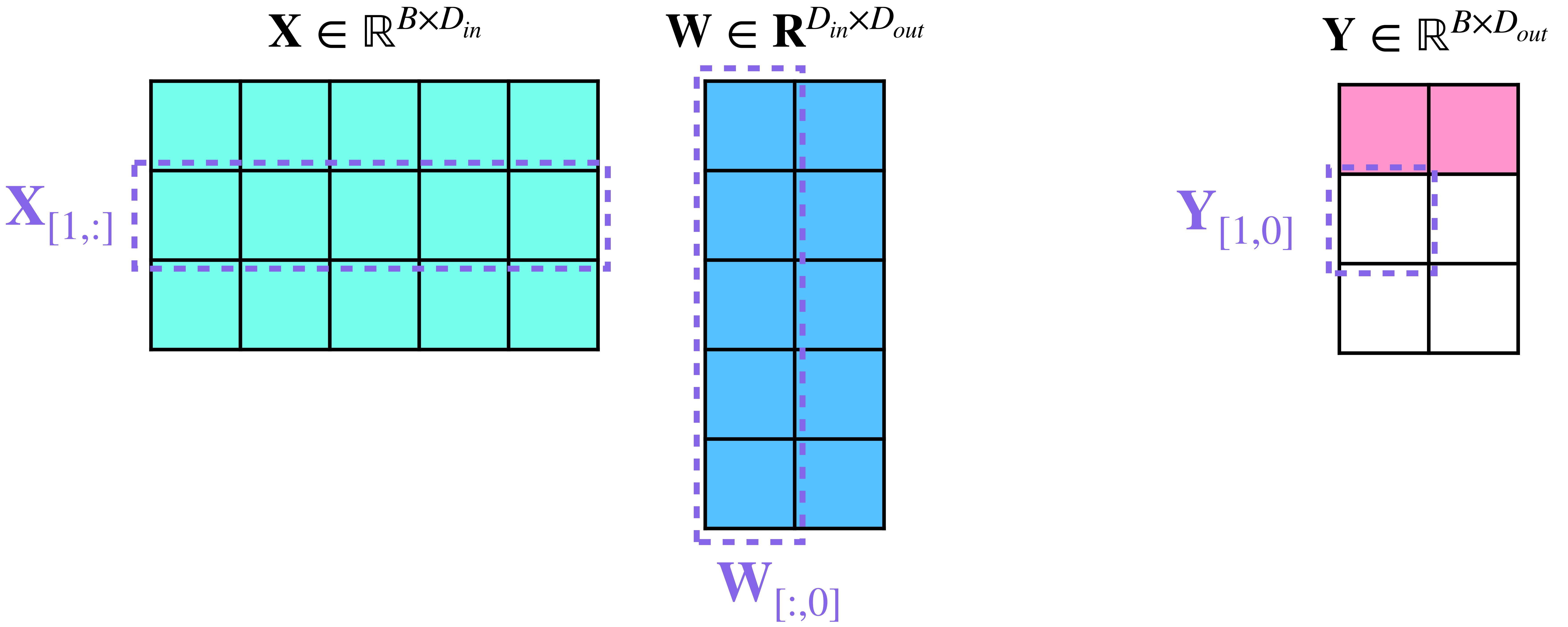
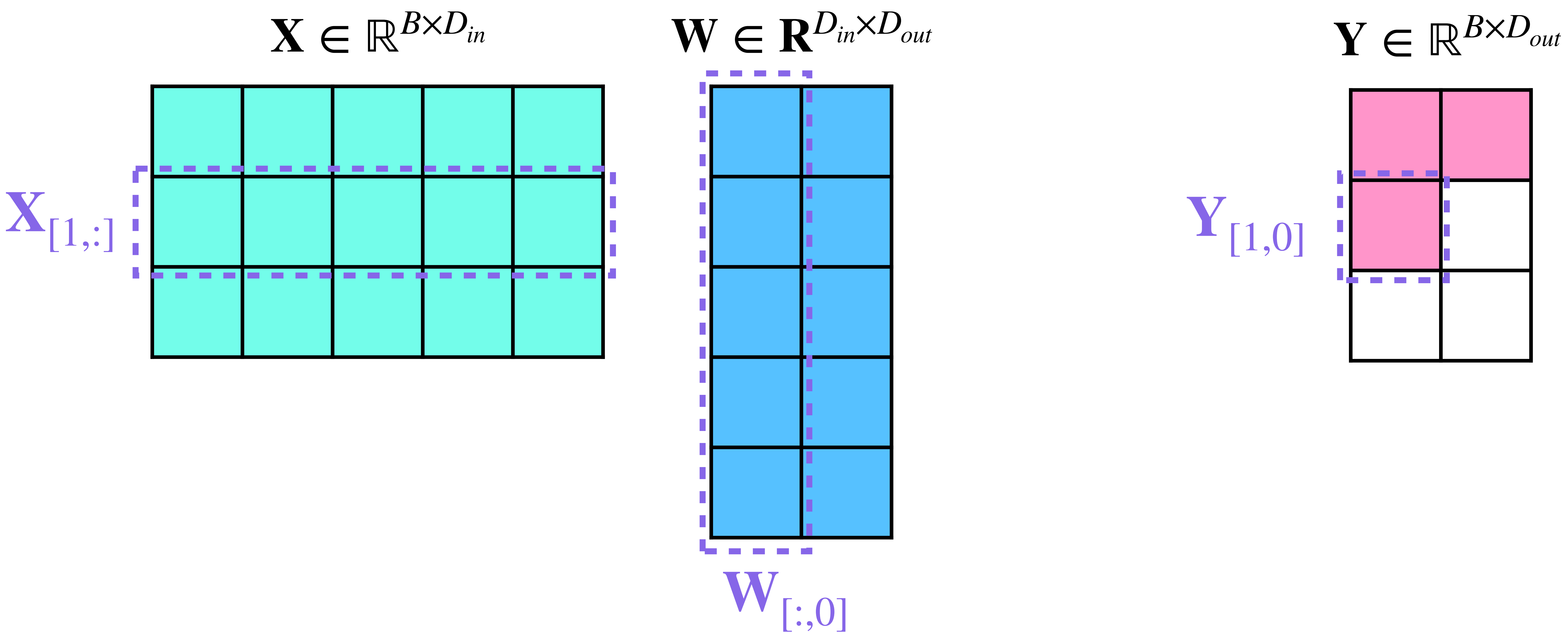


Repeat for the next prompt in the batch

A recap of matrix multiplication



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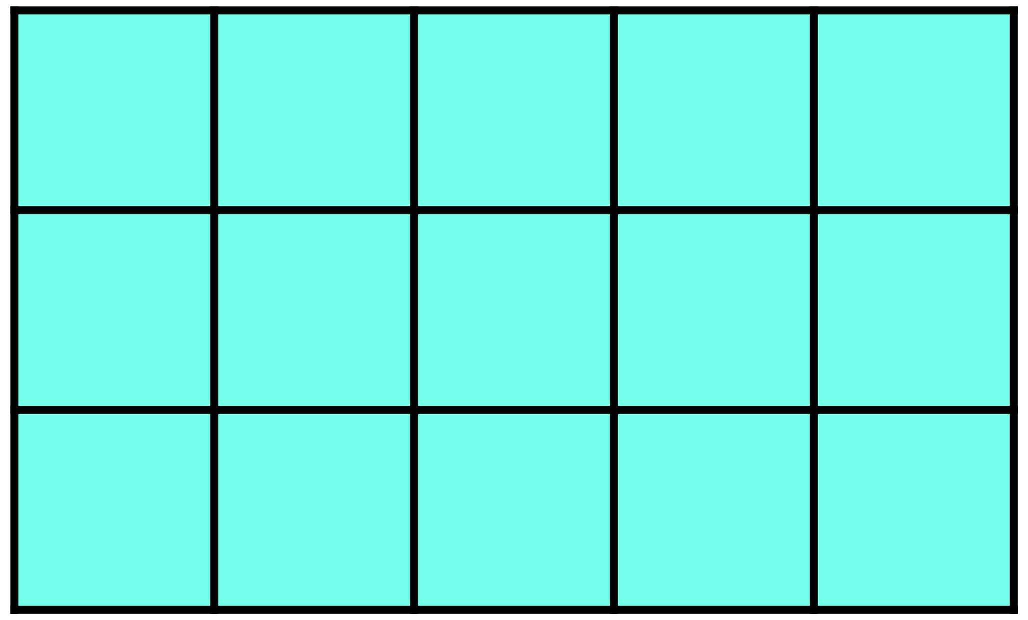
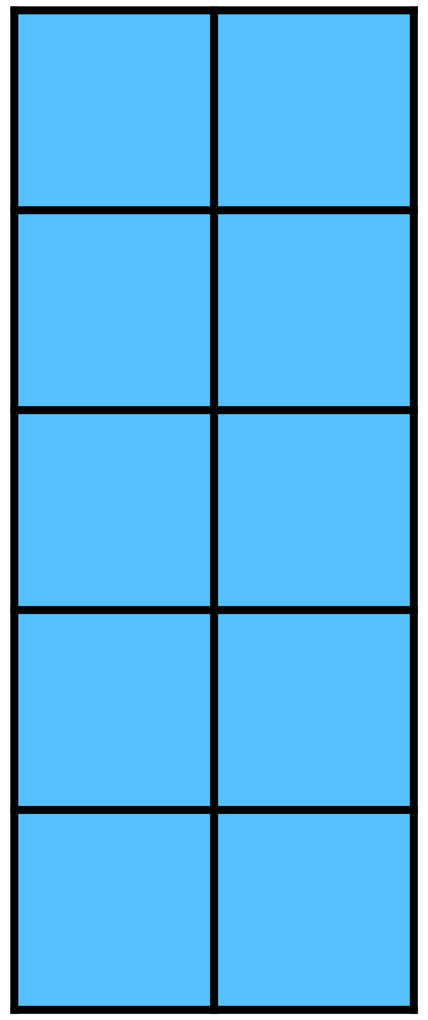
A recap of matrix multiplication

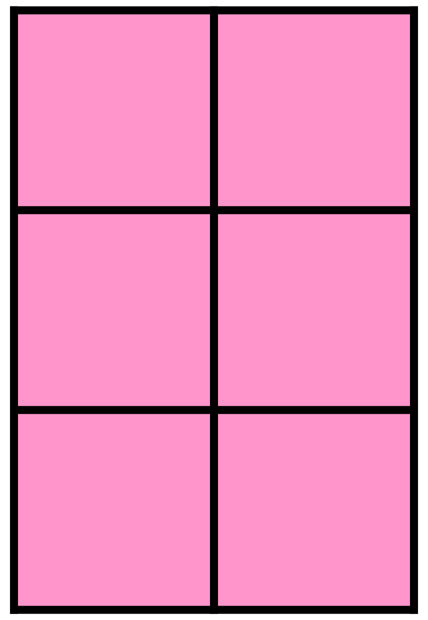
$\mathbf{X} \in \mathbb{R}^{B \times D_{in}}$



$\mathbf{W} \in \mathbf{R}^{D_{in} \times D_{out}}$

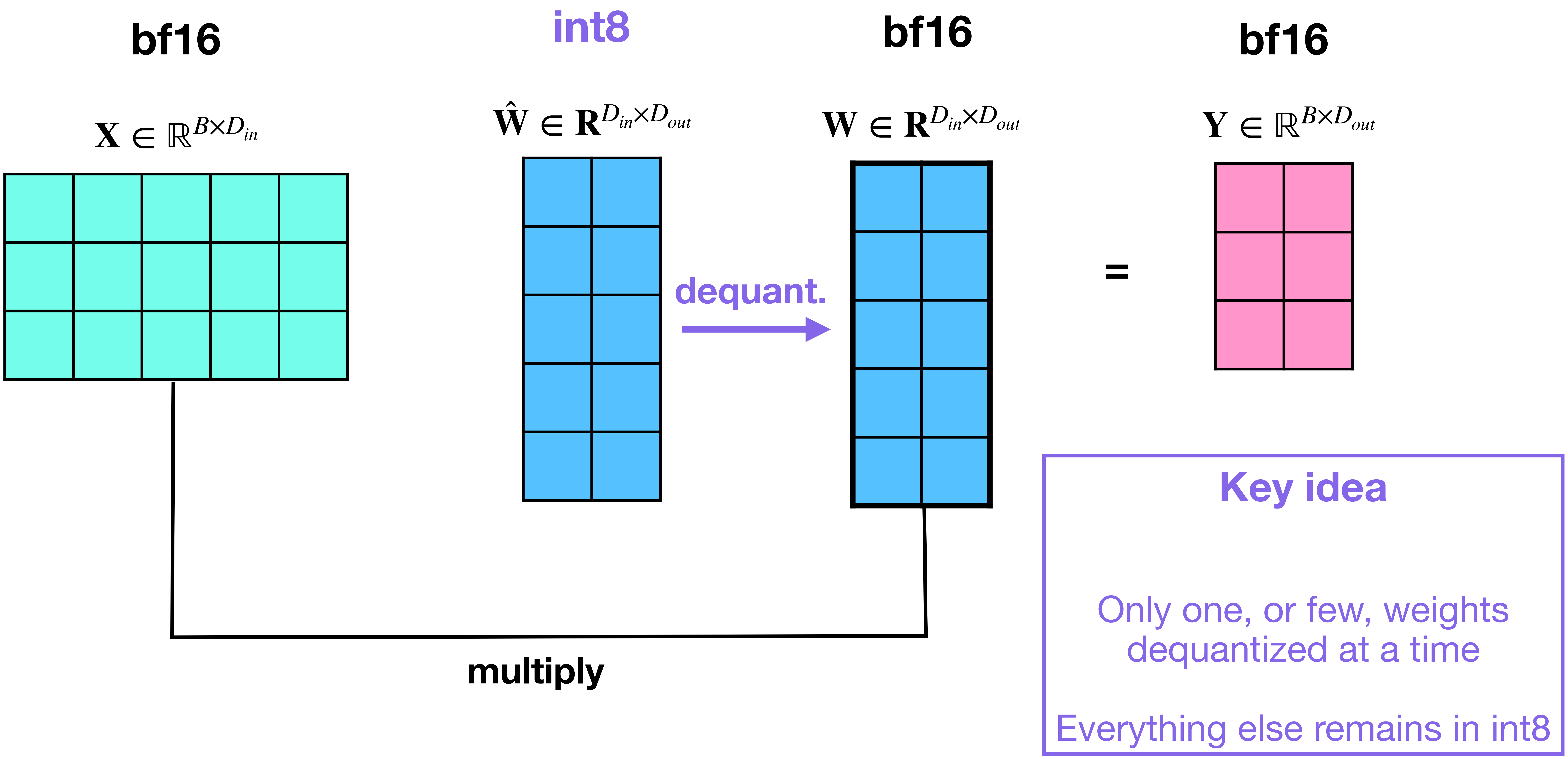


$\mathbf{Y} \in \mathbb{R}^{B \times D_{out}}$

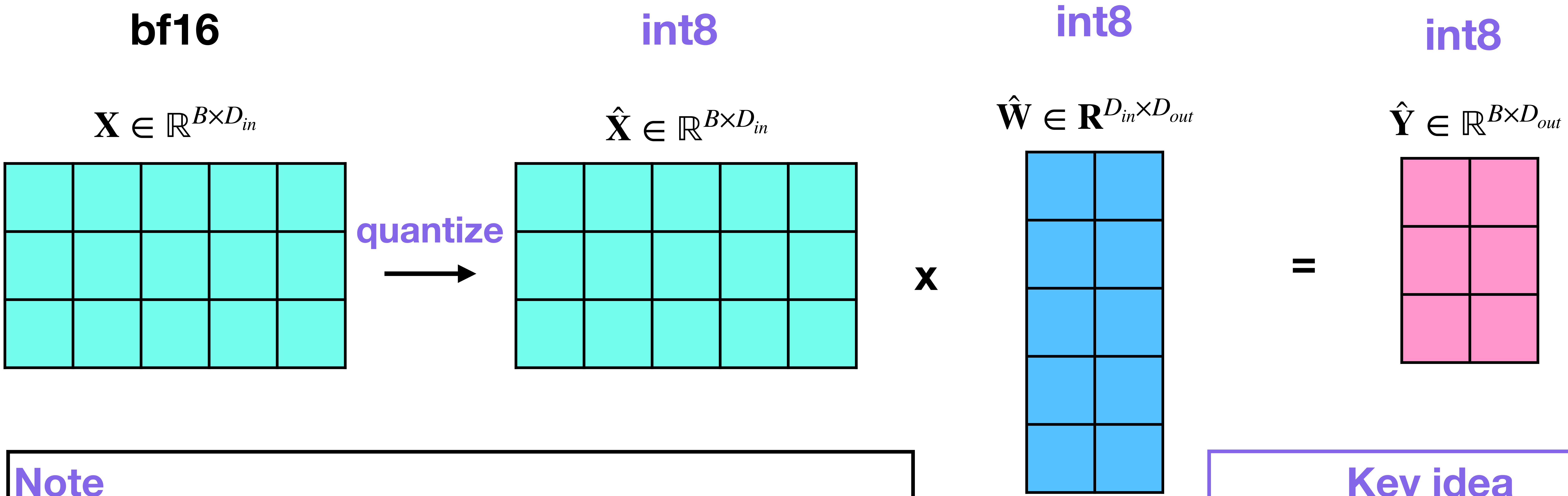


Back to quantization

Quantize weights only



Also quantize activations



Note

- ^ in $\hat{\mathbf{X}}$, $\hat{\mathbf{W}}$ and $\hat{\mathbf{Y}}$ denotes the quantized version
- $\mathbf{X}$ and $\mathbf{Y}$ are both activations
- Input of one layer L, that is, $\mathbf{X}$, is output of layer L-1

Key idea

All operations in int8

GPU can leverage <u>custom integer kernels</u>

Weight + Activation quantization

- Activation quantization adds more complications

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- Weights are fixed, that is, they do not change once we are done training
  - Quantization scale s needs to be learnt only once

$$\gamma = \frac{2^{N-1}}{\max(|\mathbf{W}|)}$$

Fixed per model

- Activations change with every new prompt
  - Need to set the quantize scale s for every new input

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Change for every prompt

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Change for every prompt

- Dynamic quantization
  - Learn the scale for every input on the fly
  - Accurate but slow (have to re-compute scale for every input)

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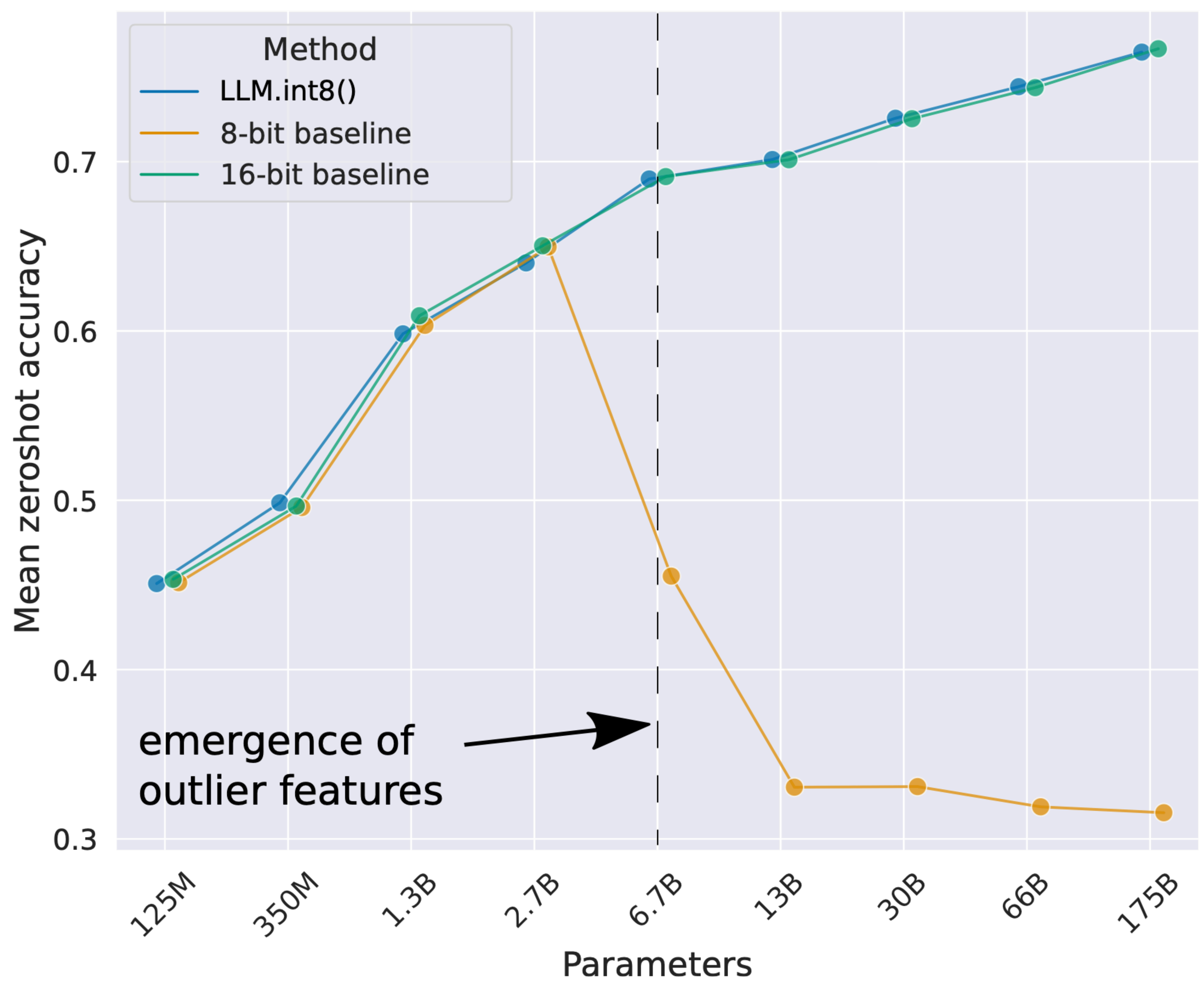
Change for every prompt

- Dynamic quantization
  - Learn the scale for every input on the fly
  - Accurate but slow (have to re-compute scale for every input)
- Static quantization
  - Use a calibration dataset (e.g., few hundred samples) to learn the scale
  - Fast but less accurate

LLM.int8() quantization

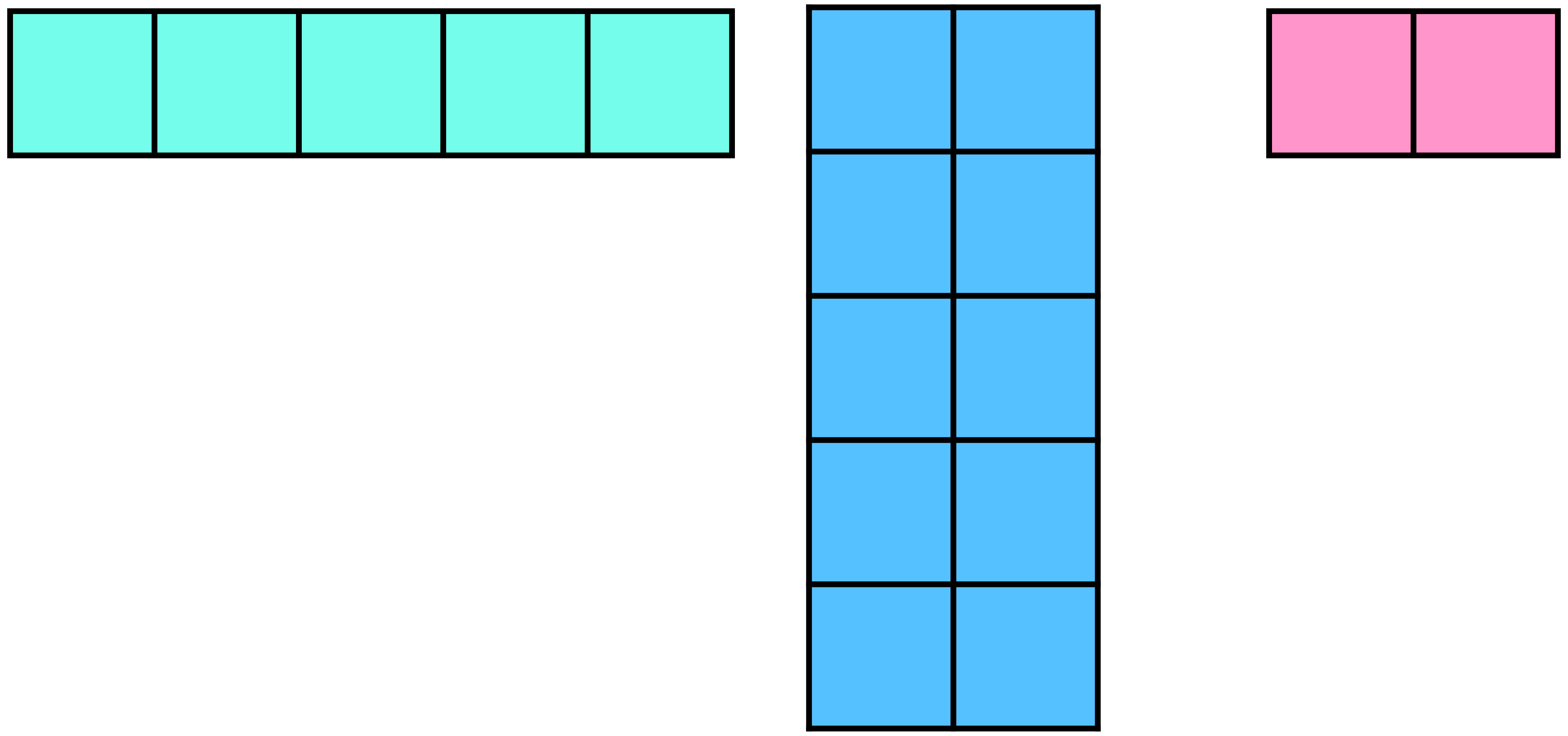
- Available in HuggingFace via the bitsandbytes library
- Outliers emerge after model size exceeds ~3B
- Luckily, outliers concentrated in few channels
- **Key idea:** Mixed precision quantization
  - Do not quantize outliers (less than 1% of the weights)
  - All other weights get quantized to int8



[Dettmers et al]

Mixed precision quantization

$\mathbf{X} \in \mathbb{R}^{D_{in}}$ $\mathbf{W} \in \mathbf{R}^{D_{in} \times D_{out}}$ $\mathbf{Y} \in \mathbb{R}^{D_{out}}$



Mixed precision quantization

LLM.int8(): Pros and cons

- Simple to implement
- Works on the fly
- Need to set the outlier threshold manually (e.g., 6)
- Paper shows that the runtime is slower than the base model
- May not work on all hardware — bitsandbytes works on GPUs only (so far)

GPTQ

- LLM.int8() performs poorly when scaled to 4 bits

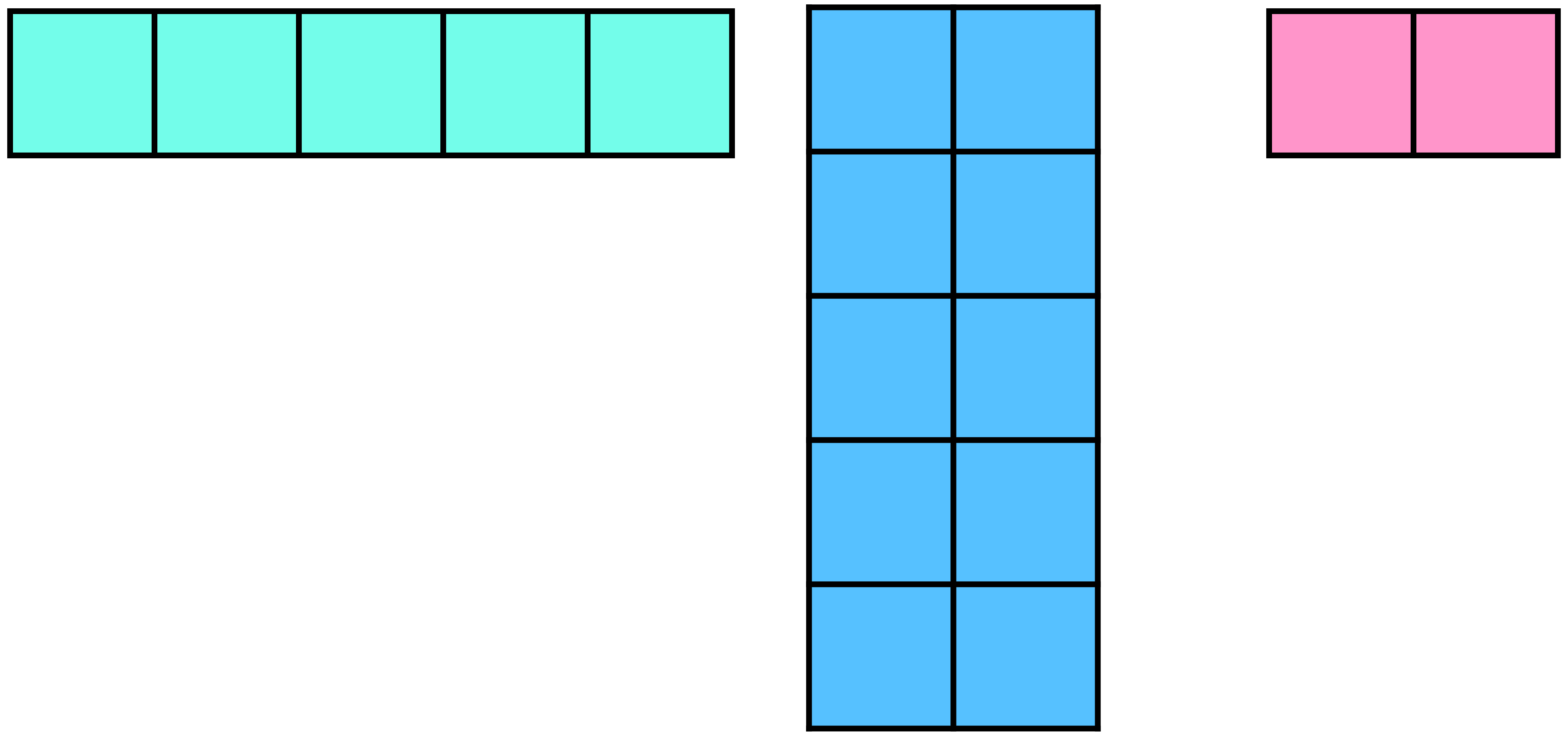
- GPTQ aims to quantize to lower than 8-bit precision, e.g., 4 or 2 bits

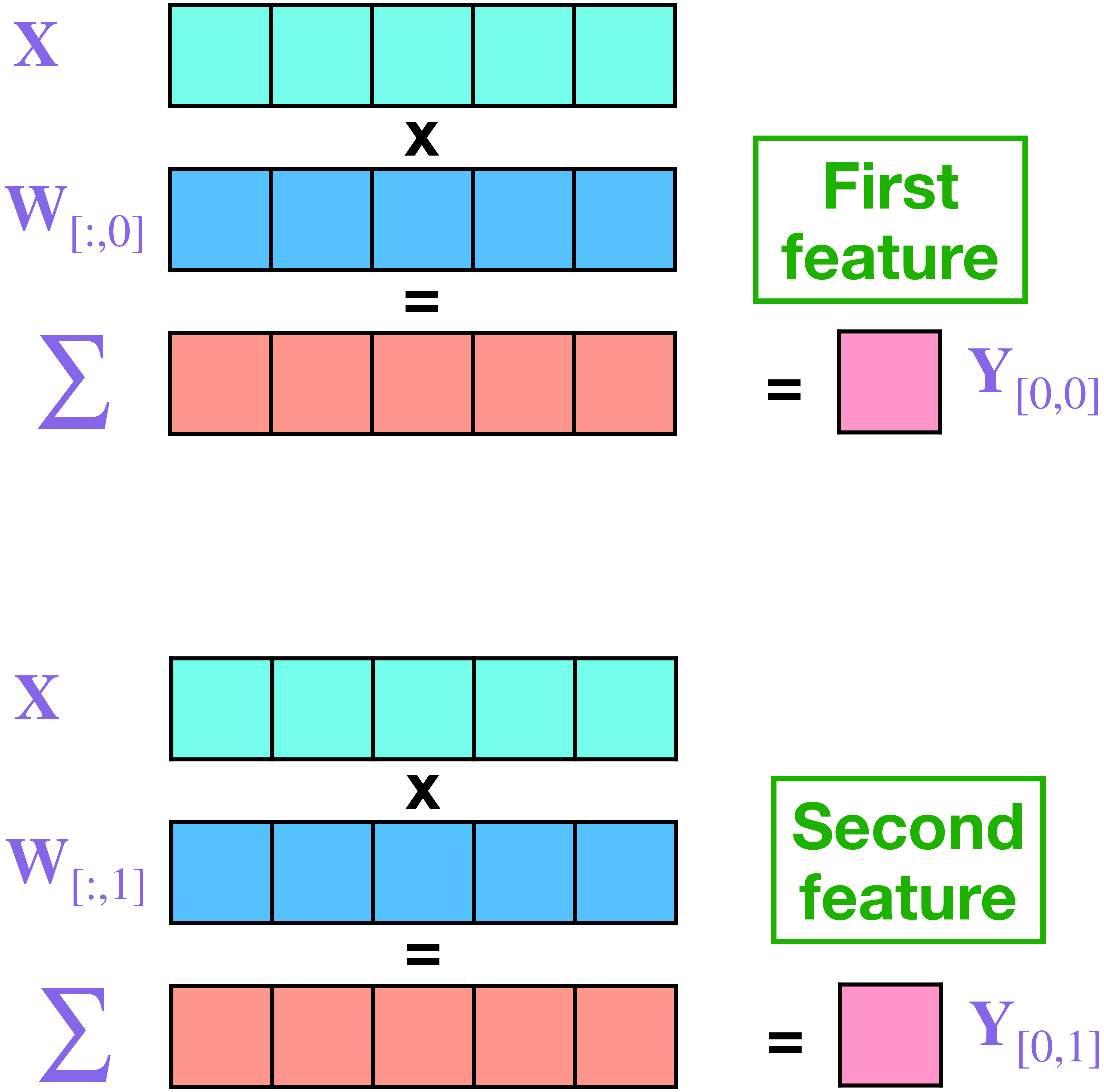
- **Key idea:** Minimize the loss between the output of quantized and unquantizd matmul

$$\hat{\mathbf{W}} = \arg\min ||\mathbf{X}\mathbf{W} - \mathbf{X}\hat{\mathbf{W}}||_2^2$$

GPTQ: Performing quantization in parallel

$\mathbf{X} \in \mathbb{R}^{D_{in}}$ $\quad$ $\mathbf{W} \in \mathbf{R}^{D_{in} \times D_{out}}$ $\quad$ $\mathbf{Y} \in \mathbb{R}^{D_{out}}$

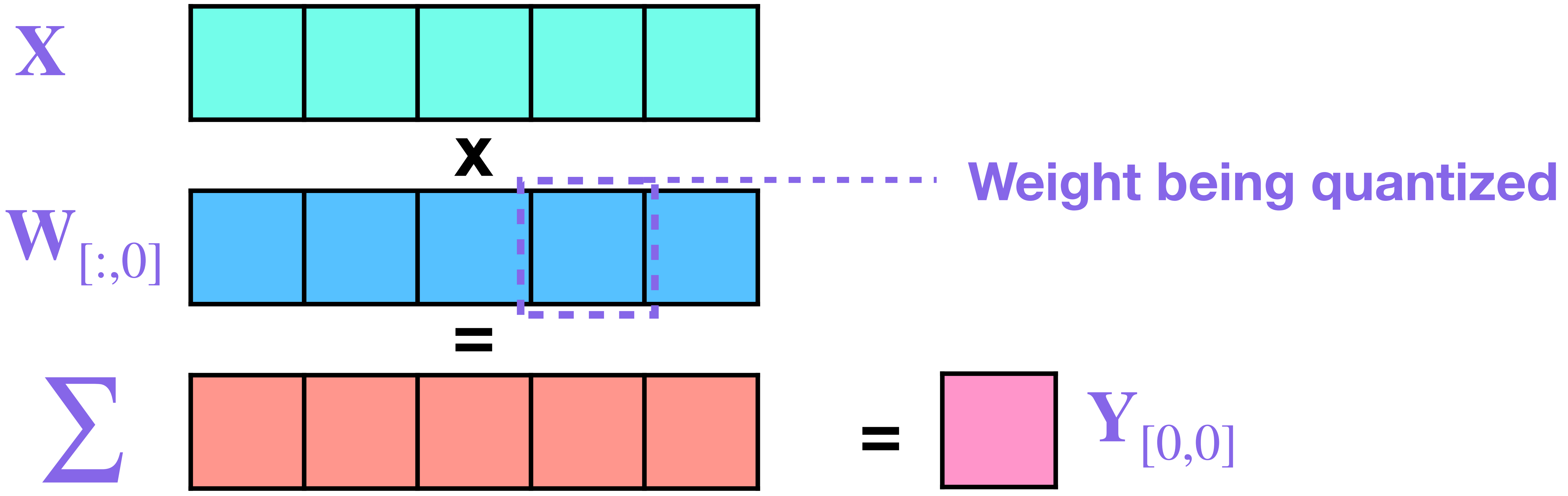




Can perform the computation for different columns **in parallel**

$$\hat{\mathbf{W}} = \arg\min ||\mathbf{X}\mathbf{W} - \mathbf{X}\hat{\mathbf{W}}||_2^2$$
$$= \arg\min \sum ||\mathbf{X}\mathbf{W}_{[:,i]} - \mathbf{X}\hat{\mathbf{W}}_{[:,i]}||_2^2$$

Iteratively quantizing weights



- Quantize this weight
- Update the unquantized weights in $\mathbf{W}_{[:,0]}$ to minimize $||\mathbf{XW} - \mathbf{X\hat{W}}||_2^2$
- Repeat for all weights in $\mathbf{W}_{[:,0]}$
- Recursively repeat for all columns

GPTQ: Pros and cons

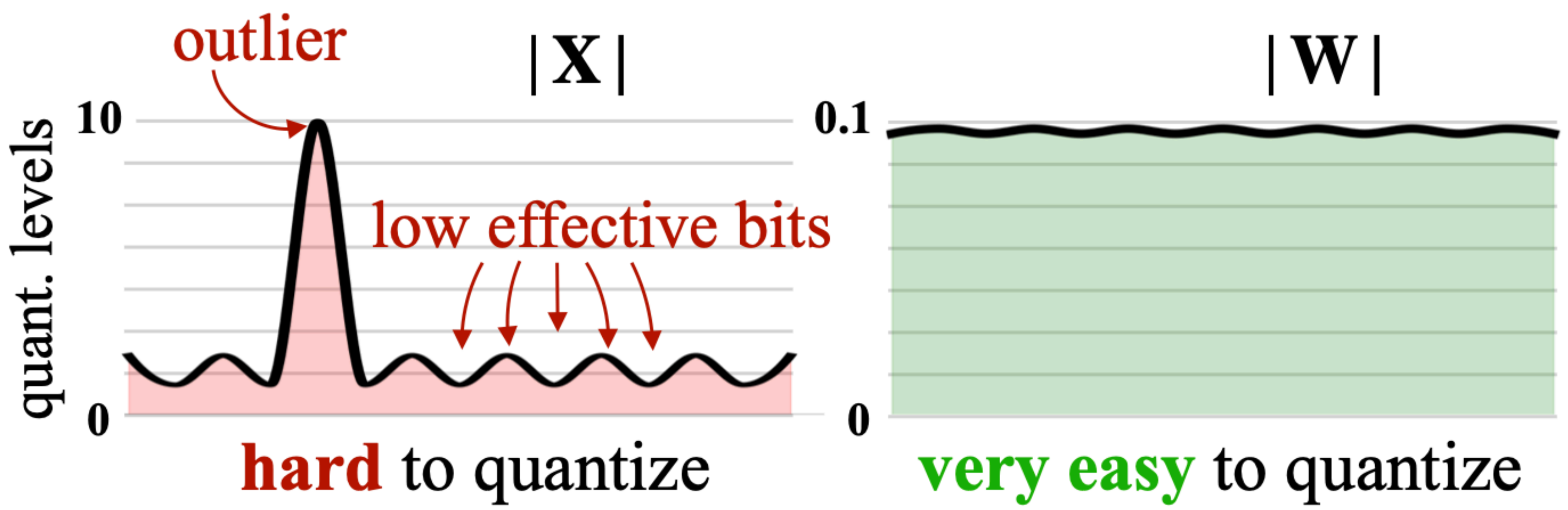
- Quantizes models very quickly by using second order optimization
- Only quantizes weights so runs on the fly
- Needs a calibration dataset
- Unlike LLM.int8(), does not need to guess the outlier values, instead, works based off the data

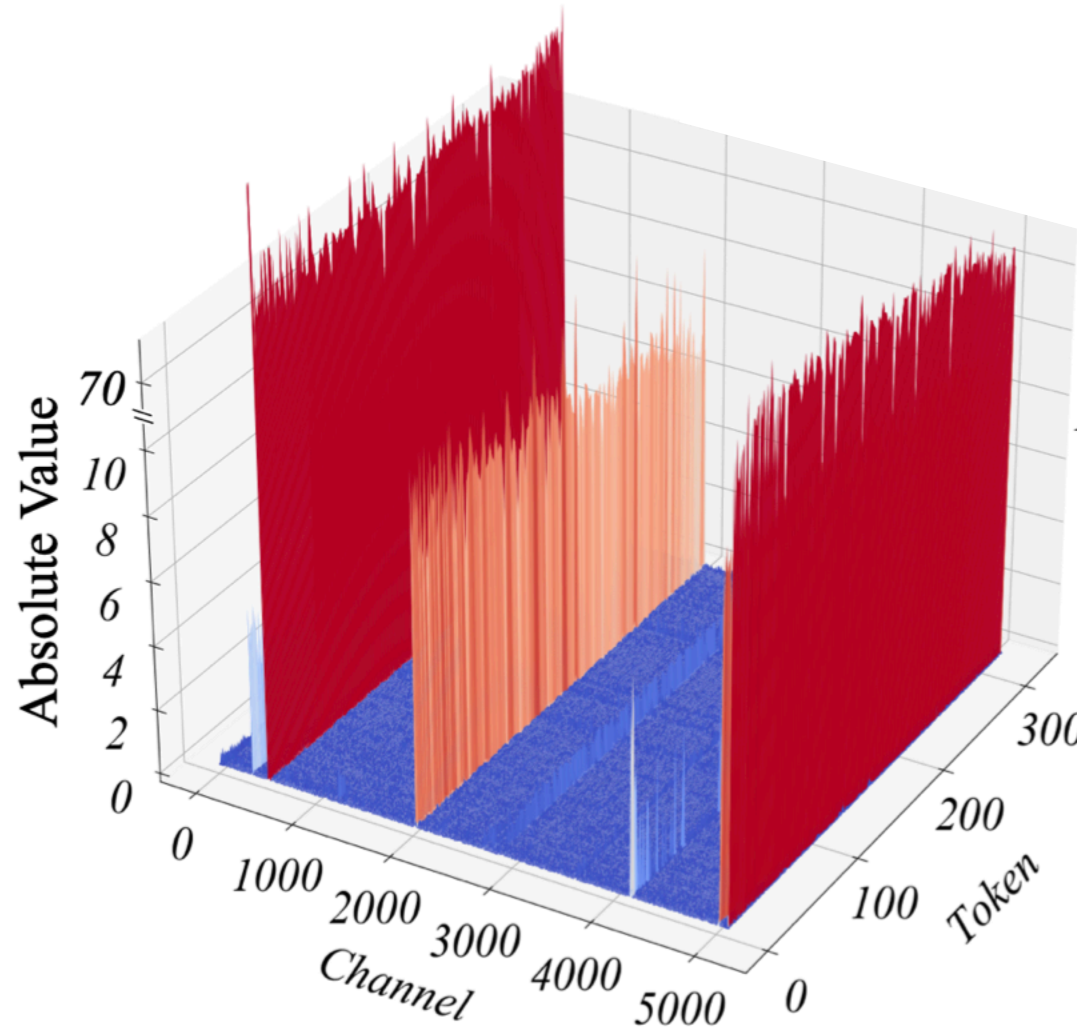
Exercise

SmoothQuant

- **Intuition:** No outliers in weights, many large outliers in activations
  - Weights: No outliers means we use the quantization range efficiently (many effective bits)
  - Activations: Large outliers means we use the quantization range inefficiently (few effective bits)

- However, outliers are concentrated in a few channels



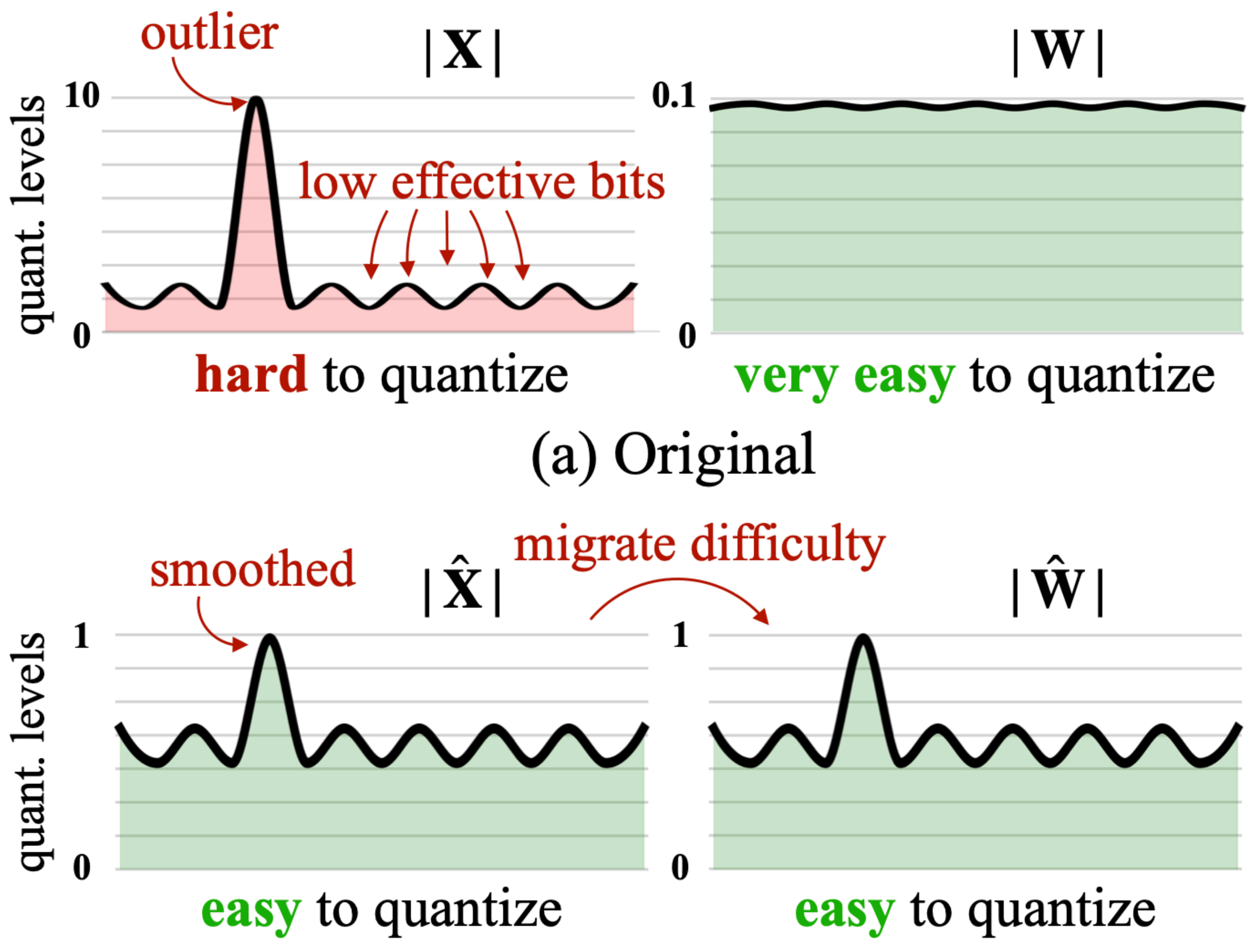


Activation (Original)
Hard to quantize

[Xiao et al.]

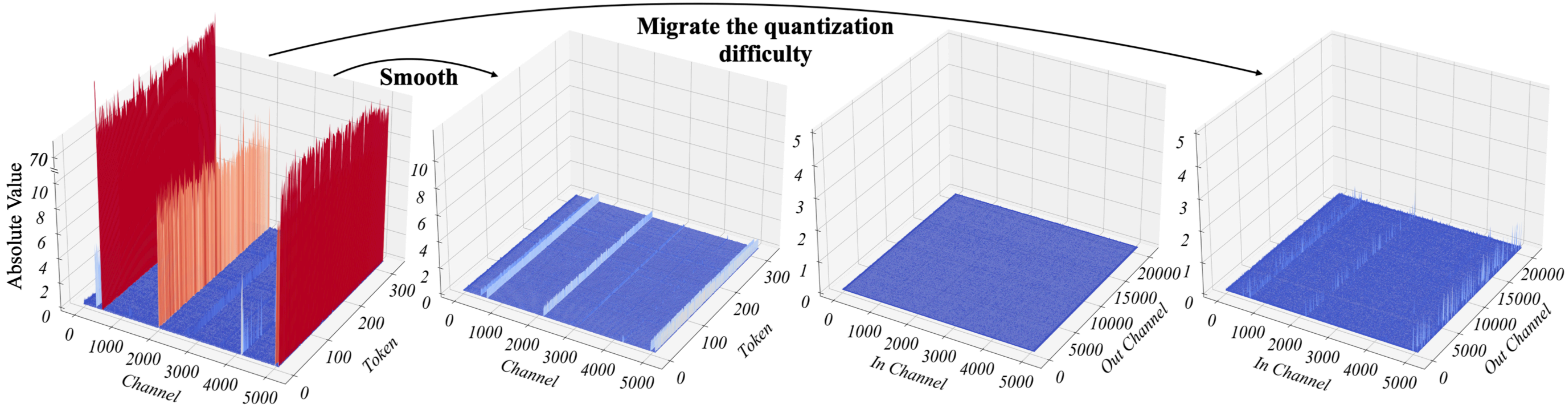
SmoothQuant

- **Key idea:** Migrating outliers between activations and weights
  - Smooth out the activations by dividing them with a channel-wise scaling term $\mathbf{s}$
  - Divide the weights by the same term to migrate the outliers
- $\bar{\mathbf{X}} = \mathbf{S}^{-1}\mathbf{X}$ with $\mathbf{X}, \mathbf{S}, \bar{\mathbf{X}} \in \mathbb{R}^{D_{in}}$ and $\mathbf{S} > 1$
- Obtain $\bar{\mathbf{W}}$ by multiplying with $\mathbf{S}$
- $\mathbf{X}\mathbf{W} = \bar{\mathbf{X}}\bar{\mathbf{W}}$
- $\mathbf{S}_j = \max(|\mathbf{X}_j|)^{\alpha} / \max(|\mathbf{W}_j|)^{1-\alpha}$
  - Pick $\alpha \in [0,1]$ based on the model / quantization
- Proceed with quantization



[Xiao et al.]

SmoothQuant



Activation (Original)
Hard to quantize

Activation (SmoothQuant)
Easy to quantize

Weight (Original)
Very easy to quantize

Weight (SmoothQuant)
Harder but still easy to quantize

[Xiao et al.]

SmoothQuant: Pros and cons

- The smoothing process done offline so faster than dynamic quantization
- Needs a calibration dataset
- Can mix and match with downstream quantization strategies

AWQ

- Somewhat of a mix of ideas discussed earlier

- **Key ideas**
  - Weights and activations interact, so quantization should be mindful of that
  - Scale weights and activations before quantization
  - Learn the quantization parameters by solving an optimization problem over the data
  - Avoid mixed precision like mix of int8 and bf16

$$\mathbf{s}^* = \arg\min_{\mathbf{s}} \mathcal{L}(\mathbf{s})$$

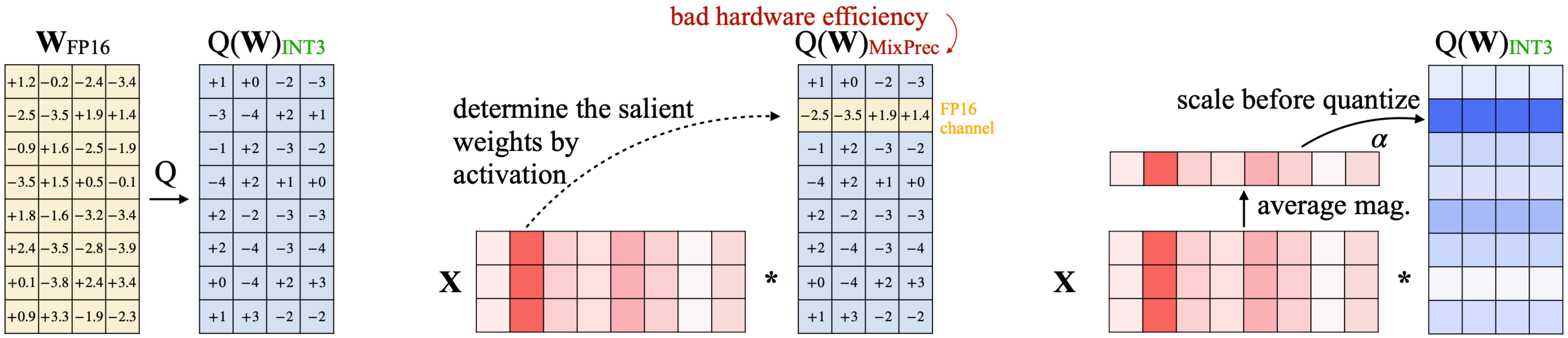
$$\mathcal{L}(\mathbf{s}) = \| \underbrace{Q}_{\text{Quantization function e.g., int4}}(\mathbf{W} \cdot \mathrm{diag}(\underbrace{\mathbf{s}}_{\text{Scaling factor (per channel)}}))(\mathrm{diag}(\mathbf{s})^{-\mathbf{1}} \cdot \mathbf{X}) - \mathbf{W}\mathbf{X} \|$$

Quantization function
e.g., int4

Scaling factor
(per channel)

[Lin et al.]

AWQ


$\mathbf{W}_{FP16}$
$Q(\mathbf{W})_{INT3}$
Q
bad hardware efficiency
$Q(\mathbf{W})_{MixPrec}$
determine the salient weights by activation
FP16 channel
X
*
scale before quantize
α
average mag.
X
*
$Q(\mathbf{W})_{INT3}$

[Lin et al.]

AWQ: Pros and cons

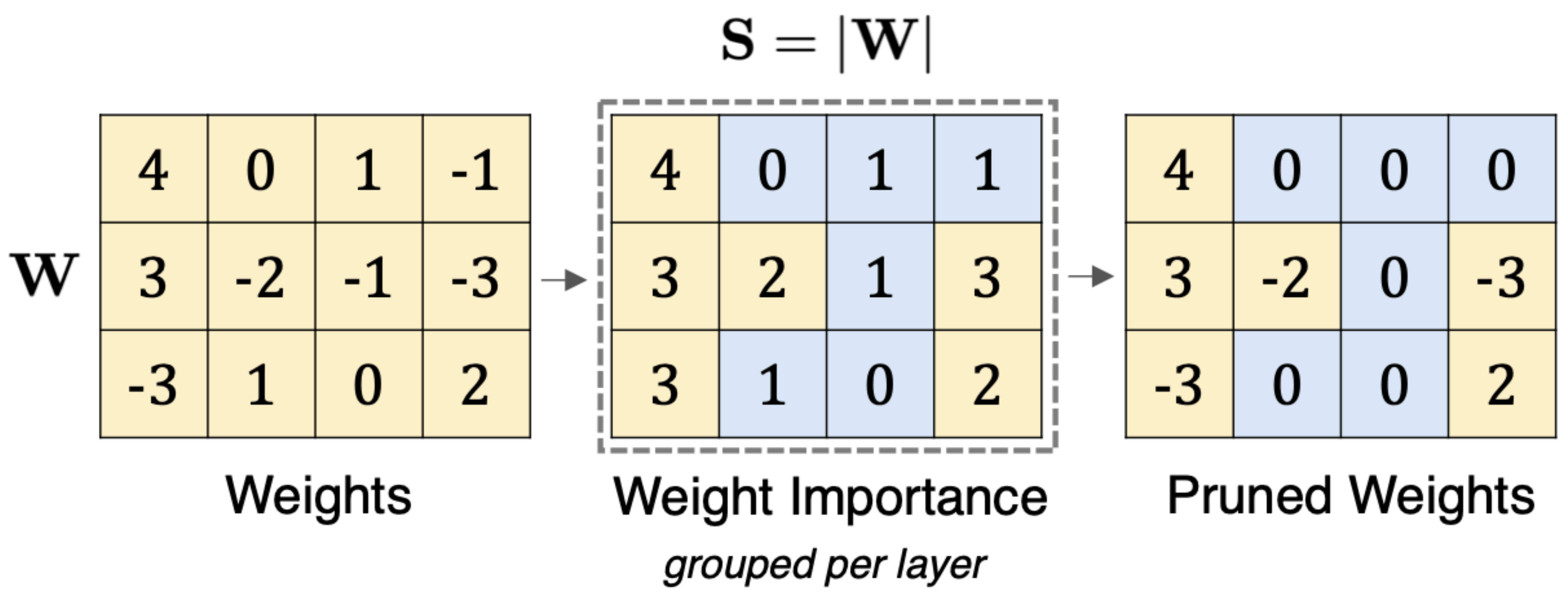
- Needs a calibration dataset
- Weight only quantization (doesn’t quantize activations)
- No mixed precision so avoids limitations of LLM.int8()
- Works well is very low precision, e.g., int4

Key considerations in quantization

- Do we need a calibration dataset?
- Is this a dynamic or static quantization strategy?
- Are only weights getting quantized or activations too?
  - Having both in int8 can leverage integer kernels to provide speedups
- Can we learn from data or do we need to make heuristic based choices?
- Does my strategy have hardware support?
  - Even int8 is not supported on all hardware
  - Some hardware does not support bf16
- Space savings and computational speed ups are two different things
  - Not all quantization methods will lead to a speedup, some will slow you down, e.g., LLM.int8()

Looking beyond quantization

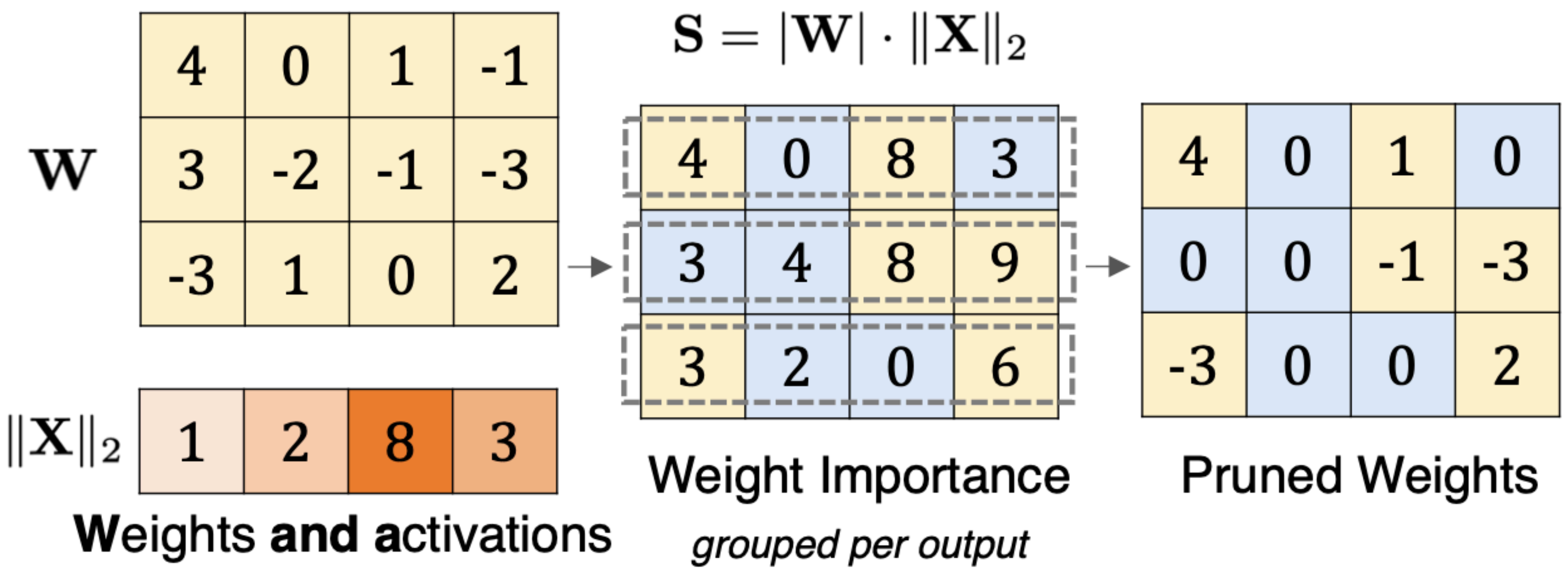
Magnitude pruning



- **Key idea:** Prune weights that are lower than a certain threshold

[Sun et al.]

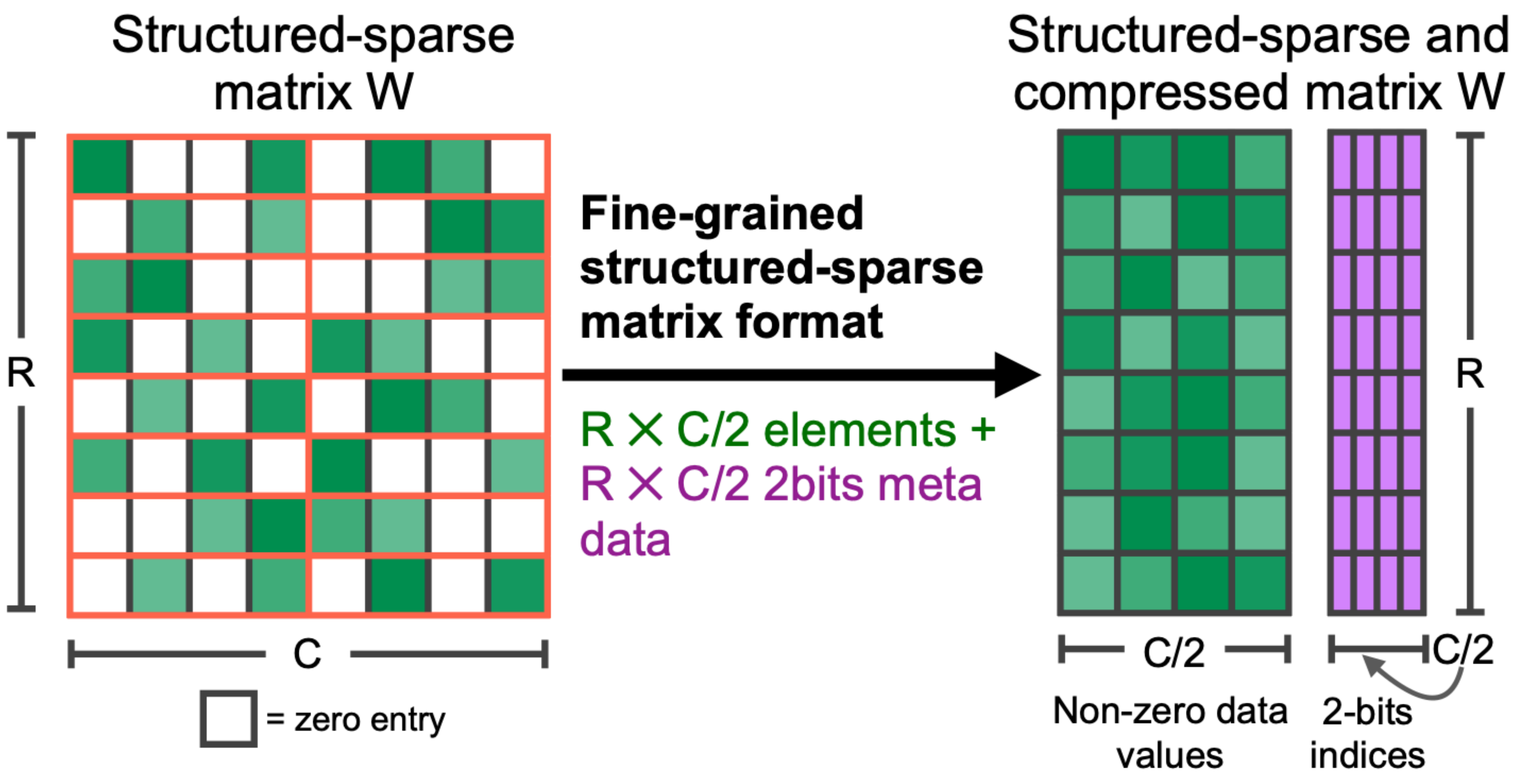
WandA (weights and activations) pruning



- **Key idea:** Prune weights that have a low importance based on the $|\mathbf{W}| \cdot ||\mathbf{X}||_2$ metric
- Similar idea to AWQ and LLM.int8(): Weight importance is determined by the activations too

[Sun et al.]

Structured pruning



- **Key idea:** Fixed ratio of weights (e.g., 2:4) is pruned between each structure
- Can leverage hardware acceleration

[Mishra et al.]

Exercise

References

- Making LLMs lighter with AutoGPTQ and transformers
- Introduction to Weight Quantization
- A Gentle Introduction to 8-bit Matrix Multiplication for transformers at scale using Hugging Face Transformers, Accelerate and bitsandbytes
- A Visual Guide to Quantization
- SmoothQuant: Accurate and Efficient Post-Training Quantization for Large Language Models
- GPTQ: Accurate Post-Training Quantization for Generative Pre-trained Transformers
- AWQ: Activation-aware Weight Quantization for LLM Compression and Acceleration
- LLM.int8(): 8-bit Matrix Multiplication for Transformers at Scale
- A Simple and Effective Pruning Approach for Large Language Models
- Accelerating Sparse Deep Neural Networks
- MIT 6.5940: TinyML and Efficient Deep Learning Computing